

— AN OPLURIX PRODUCT · VERSION 1.0 · EN

Thermal Data *Translator*

Convert drone thermal imagery into validated wildlife detection records, habitat quality maps, and analysis-ready files — from raw RJPEG output to MRV-compatible biodiversity data.

BUILT FOR

Drone-Equipped Research Teams

Wildlife Survey Coordinators

Conservation NGO Tech Leads

ERIS / MRV Data Managers

THE TRANSFORMATION

From "I have 800 thermal image files and no idea how to turn them into detection records" to "I have a validated wildlife detection dataset with GPS, species confidence flags, and habitat quality metrics — ready for occupancy modelling or biodiversity scoring."

HOW TO USE THIS PRODUCT

Read Sections 1–3 once. This is the most technical OPLURIX product — it assumes you have flown a thermal drone survey and have the image files. Sections 4–6 are the working workflow. The AI Prompt Pack (Section 7) and Code Templates (Section 9) are your day-to-day tools. The Species Detection Guide (Section 8) is your field-specific reference for Madagascar and tropical forest contexts.

Contents: Core Promise · Pain Points · Desired Outcomes · 7-Step Processing Pipeline · Thermal Image Interpretation Guide · Flight Planning for Data Quality · AI Prompt Pack · Species Detection Guide (Madagascar) · Python/R Code Templates · Detection Record Schema · Habitat Quality Scoring · ERIS Integration · QC Framework · Quick Start

SECTION 1

The Core Promise

Thermal drone surveys produce data at a rate that quickly overwhelms manual processing. A single 25-minute thermal flight at Ihofa Forest generates between 300 and 800 image files, depending on flight speed and trigger interval. Each image must be reviewed, detections must be validated, false positives must be removed, GPS coordinates must be extracted and verified, and the results must be entered into a structured detection dataset. Multiply this by a week of field surveys and you have a data processing challenge that can take longer than the survey itself — if you do not have a system.

This guide gives you that system. It is built on the actual workflow used for the thermal drone lemur surveys presented at IPS Congress 2025 — the proof-of-concept that underpins the ERIS methodology. It covers the full pipeline from raw `.RJPEG` or `.TIFF` thermal output to a validated, GPS-tagged, confidence-flagged wildlife detection dataset that is directly compatible with the OPLURIX Biodiversity Data Harmonizer and the ERIS scoring framework.

THE TRANSFORMATION

Before: "I have 800 thermal image files from last week's survey and no idea how to turn them into detection records."

After: "I have a validated wildlife detection dataset with GPS coordinates, species confidence flags, thermal signature metadata, and habitat quality metrics — ready for occupancy modelling, biodiversity scoring, or ERIS input."

What this guide covers: Thermal image file formats and extraction, flight parameters affecting data quality, manual and AI-assisted detection workflows, false positive identification and removal, species confidence classification, GPS coordinate extraction and validation, habitat quality assessment from

thermal data, detection record schema, YOLO-based detection model guidance, and integration with the ERIS biodiversity scoring system.

This guide does not cover: How to fly a drone, how to obtain flight permits, basic drone operation, or the physics of thermography. It assumes you have a completed thermal survey dataset and want to extract scientific value from it.

10 Real Pain Points

01 You cannot tell a warm animal from a warm rock.

Thermal signatures are temperature differentials, not species identifications. Sun-warmed rocks, decomposing vegetation, and reflective surfaces all produce warm signatures that look like animals in thermal imagery. Without a systematic false positive protocol, your detection count is inflated and your dataset is scientifically unusable.

02 The same individual appears in multiple consecutive frames.

A thermal drone flying at 5 m/s with a 1-second image trigger interval takes a new image every 5 metres. A 2-metre animal can appear in 3–5 consecutive frames. Without a frame-merging protocol, you count one animal five times.

03 You cannot identify species from thermal imagery alone.

Thermal cameras detect heat, not morphology. A small lemur and a large gecko at the same distance produce different-sized signatures — but size alone cannot reliably distinguish species in dense canopy. Species classification requires a combination of thermal signature shape, size estimation, behaviour, habitat context, and cross-validation with concurrent audio or RGB data.

04 Dense canopy blocks thermal signatures completely.

Under a closed forest canopy, the thermal camera sees canopy surface temperature — not the animals underneath. Detection rates in closed canopy are dramatically lower than in open canopy or edge habitats, but this difference is a detection artefact, not a real ecological signal. It must be accounted for in any analysis.

05 GPS metadata in thermal image files is often inaccurate or missing.

Consumer-grade drones (including the Mavic 3T) embed GPS coordinates in image EXIF data from the aircraft's onboard GPS — not RTK-grade positioning. The typical accuracy is ± 2 –5 metres in open sky, degrading to ± 10 –20 metres in forested terrain where satellite acquisition is poor. Understanding and documenting this uncertainty is essential for spatial analysis.

06 You have no system for consistent image review across multiple observers.

When two people review the same thermal image sequence, they will flag different detections. Without a standardised review protocol and inter-observer calibration, your detection dataset reflects the reviewer as much as the actual wildlife.

07 The thermal image file format is not directly importable into standard analysis tools.

DJI thermal cameras produce `.RJPEG` files — a proprietary format that contains both a standard JPEG preview and raw radiometric temperature data. Standard image viewers show the JPEG. The radiometric data, which is required for temperature-based analysis, requires specific extraction tools. Most researchers never access the raw temperature data at all.

08 Time of survey profoundly affects detection probability — but nobody documents it systematically.

Thermal contrast between an endothermic animal and the environment is highest in the hours before dawn, when the environment is coldest. At midday in a tropical forest, thermal contrast may be so low that detection is near-impossible. This is not a equipment failure — it is a survey design issue. But without systematic documentation of survey time and ambient temperature, the detection probability varies dramatically across a dataset with no way to model it.

09 YOLO models trained on other ecosystems perform poorly on your species.

A YOLO model trained on African savanna wildlife will not reliably detect Malagasy forest lemurs in dense canopy thermal imagery. Models must be trained or fine-tuned on imagery from your specific ecosystem, your specific species, and your specific flight parameters. This guide explains what you need to build and validate a context-specific model.

10 You cannot connect thermal detections to the ERIS scoring framework without a structured output format.

The ERIS ecological integrity scoring system requires detection data in a specific schema — with confidence flags, GPS coordinates, species classification confidence, canopy cover estimates, and effort metadata all attached to each record. Raw thermal imagery does not produce this format. This guide ends with the exact output schema needed for ERIS integration.

SECTION 3

10 Desired Outcomes

#	Outcome
01	Process a full day's thermal survey imagery into a validated detection dataset in one working session.
02	Distinguish genuine animal thermal signatures from false positives systematically — with documented criteria.
03	Merge consecutive-frame detections of the same individual into single validated records.
04	Classify detections by species confidence level (HIGH / MEDIUM / LOW / UNIDENTIFIED) with documented criteria.
05	Extract and validate GPS coordinates from thermal image EXIF data.
06	Document detection effort — survey time, altitude, canopy cover, ambient temperature — as a covariable for detection probability modelling.
07	Build or fine-tune a YOLO detection model appropriate for your target species and ecosystem.
08	Generate habitat quality thermal maps that characterise canopy cover and thermal heterogeneity.
09	Produce an output detection dataset directly compatible with the ERIS scoring schema.
10	Validate the detection dataset through a three-check quality framework before any ecological analysis.

The 7-Step Thermal Processing Pipeline

From raw image files to validated detection records. Every step has a documented output. Steps 1–3 are image-level work. Steps 4–7 build the scientific dataset.

FILE ORGANISATION AND METADATA EXTRACTION

Before reviewing a single image, organise the files and extract the metadata. This step defines the structure every subsequent step works within.

FILE ORGANISATION PROTOCOL

- One folder per flight. Naming: `[SiteCode]_[Date]_[FlightNum]` — e.g., `IH0FA_20260315_F01`
- Do not rename individual image files. The original filename contains the sequence number used for consecutive-frame merging.
- Copy all files to working storage before processing. Never work from the original memory card.
- For each flight folder: create a `_metadata.csv` file with one row per image, extracting: filename, GPS latitude, GPS longitude, GPS altitude, image timestamp, camera model, thermal palette setting.

Use the Python EXIF extraction template in Section 9 (Prompt 10) to batch-extract metadata from all files in a flight folder. The `exiftool` command-line tool is the most reliable option for DJI thermal files.

CRITICAL: RJPEG VS. TIFF FILE HANDLING

DJI thermal cameras (Mavic 3T, H30T, XT2) produce `.RJPEG` files. These contain a standard JPEG preview (visible to any image viewer) and embedded radiometric data (temperature per pixel). For wildlife detection, the JPEG preview is usually sufficient. For temperature-based analysis (habitat thermal mapping, thermal habitat quality scoring for ERIS), the radiometric data must be extracted separately using DJI's `dji_thermal_sdk` or the open-source `thermal_parser` Python library.

Never delete RJPEG files after extracting the JPEG preview — the radiometric data is embedded in the original file and cannot be recovered once the file is deleted.

Output: Organised flight folders + metadata CSV per flight.

THERMAL PALETTE NORMALISATION

Before reviewing images for detections, ensure all images in a flight use the same visual presentation. Mixed palettes (some images in Ironbow, some in Rainbow) make cross-image comparison impossible and dramatically increase reviewer error rates.

RECOMMENDED PALETTE FOR WILDLIFE DETECTION

Ironbow (Iron/Hot): Warm objects appear orange-white, cold objects appear dark blue-black. High contrast between endothermic animals and cool vegetation. This is the standard palette for wildlife detection in forested environments.



Cold (dark)

→ Ironbow palette →

Hot (white/yellow)

For habitat mapping: Rainbow palette provides more gradation across the temperature range. Use for thermal heterogeneity analysis, not for detection review.

Output: All images in consistent Ironbow palette. Note the temperature range displayed (e.g., 15°C–40°C) in the flight metadata — this affects how detections appear.

INITIAL DETECTION REVIEW — FLAGGING

Review every image in the flight sequence and flag images containing candidate detections. Do not attempt species identification at this stage. You are answering one question only: "Is there a thermal signature in this image that warrants closer review?"

DETECTION FLAGGING CRITERIA

- A blob-shaped warm signature distinctly warmer than the surrounding canopy surface — not a gradual warm area
- A signature with a shape consistent with a vertebrate body (elongated, roughly symmetrical, with a definable edge)
- A signature that changes position across consecutive frames (moving animals)
- A signature that appears and disappears across frames (animal emerging from and returning to canopy cover)

Record the image filename of every flagged frame. The fastest review method is fast-scroll through images at 4–6 frames per second, pausing to flag anything that meets the criteria. A 500-image flight can be reviewed for flagging in 20–30 minutes at this pace.

Output: A flagged image list — filenames of all images containing candidate detections. Typically 5–15% of total images in a forested tropical survey.

FALSE POSITIVE REMOVAL

For every flagged image, apply the false positive decision framework. This is the most important step for data quality. A detection dataset with 30% false positives is not usable for any rigorous analysis.

THE SIX MOST COMMON FALSE POSITIVES IN TROPICAL FOREST THERMAL SURVEYS

- **Sun-warmed rock or boulder:** Appears very bright (hotter than expected ambient). Diagnostic: present in same frame across multiple consecutive flights; does not move; edges are sharp geometric shapes.
- **Decomposing vegetation:** Mounds of decaying organic matter produce significant heat. Diagnostic: diffuse warm area, irregular edges, no movement, typically on the ground.
- **Termite mounds:** Extremely hot signatures. Diagnostic: stationary, very high temperature relative to surroundings, spherical or conical shape, consistent across multiple days.
- **Reflective surfaces:** Metal objects, water, and wet leaves can reflect solar radiation and appear warm. Diagnostic: signature too bright (clipping the temperature scale), sharp geometric edges.
- **Thermal bleed from canopy gap:** Solar heating of the ground visible through a gap in the canopy creates a warm patch that looks like an animal on the ground. Diagnostic: signature corresponds to a canopy gap visible in simultaneous RGB image; stationary; uniform warming pattern.
- **Aircraft thermal artefacts:** Hot pixels or banding from camera sensor artefacts. Diagnostic: same pixel position in every image of the same flight; not correlated with ground features.

Output: Flagged image list with false positives removed. Each removal documented with the false positive category applied.

CONSECUTIVE-FRAME MERGING

Merge all consecutive-frame observations of the same individual into a single detection record. This is the most common counting error in thermal drone surveys.

CONSECUTIVE-FRAME MERGING RULES

- Group flagged images that are sequential in frame number (no gap larger than 3 frames between them)
- For each group: verify that the signature position is consistent with the drone's flight path (the signature moves through the frame as the aircraft passes)
- For each group: count one individual, not one image
- The detection GPS coordinate is taken from the image where the animal is closest to the nadir (directly below the aircraft) — the most accurate GPS estimate for the animal's position
- If two separate signatures appear in the same frame sequence and maintain consistent separation: count as two individuals

FRAME GAP RULE

A gap of 1–3 frames (1–3 seconds at standard trigger rate) between observations of what appears to be the same animal can occur when the animal briefly moves under canopy between images. A gap larger than 5 frames should be treated as a potential second individual unless the GPS positions are consistent with continuous movement.

Output: One row per unique individual detection. Frame group documented in the `source_frames` column.

SPECIES CLASSIFICATION AND CONFIDENCE ASSIGNMENT

Classify each validated detection to the highest taxonomic level that the data supports with confidence. Do not overstate species identification from thermal data alone.

Confidence level	Criteria	Taxonomic resolution
HIGH	Signature size, shape, and behaviour clearly diagnostic. Simultaneous audio or visual confirmation available. OR YOLO model confidence ≥ 0.85 on a validated model.	Species level: e.g., <i>Indri indri</i>
MEDIUM	Signature size and habitat consistent with one species. No alternative species equally plausible. No simultaneous confirmation.	Species level with "cf." flag: e.g., cf. <i>Indri indri</i>
LOW	Signature consistent with a taxonomic group but size or behaviour ambiguous. Multiple species equally plausible.	Genus or family: e.g., <i>Hapalemur</i> sp. or LEMUR_INDET
UNIDENTIFIED	Warm signature clearly biological but no identification possible — canopy obstruction, too small, too brief.	Thermal detection: e.g., WARM_SIG_MAMMAL or WARM_SIG_REPTILE

Output: Species code, species confidence level, and classification basis documented for each detection.

GPS EXTRACTION, VALIDATION, AND EFFORT DOCUMENTATION

Extract GPS coordinates from image EXIF data. Validate against the study area. Document all flight parameters as survey effort covariables.

EFFORT COVARIABLES — RECORD FOR EVERY FLIGHT

- **Flight time:** Start and end time (UTC). Used to calculate ambient temperature differential.
- **Flight altitude AGL:** Above-ground level in metres. Affects sensor footprint size and detection distance.
- **Flight speed:** m/s. Affects consecutive-frame overlap and detection probability.
- **Image trigger interval:** Seconds between frames.
- **Ambient temperature:** °C at survey time. The single most important covariable for thermal detection probability.
- **Canopy cover:** Estimated percentage for each transect pass. Can be derived from RGB imagery or from the thermal image itself (proportion of frame with warm canopy surface).
- **Wind speed:** m/s. Above 10 m/s, detection probability drops significantly as animals shelter.
- **GPS accuracy estimate:** From EXIF or DJI flight log. Typical Mavic 3T: $\pm 2\text{--}5\text{m}$ in open, $\pm 10\text{--}20\text{m}$ under canopy.

Output: Detection dataset with validated GPS + flight effort table. Ready for analysis.

Thermal Image Interpretation Guide

How to read thermal imagery for wildlife detection. Built from the Mavic 3T field experience at Ihofa Forest, but applicable to any wildlife thermal drone survey in tropical forest.

Understanding What the Camera Sees

A thermal camera measures the radiant temperature of the surface it can see — not the air temperature and not the temperature of objects obscured by other objects. In a forest survey, this means:

- The camera primarily sees the **canopy surface** when flying above a closed forest.
- Animals are only visible when they are **at or above the canopy surface**, or in a **gap in the canopy** that allows a sightline to the animal.
- The canopy surface temperature varies across the image — sun-facing leaves are warmer than shaded leaves, creating a background thermal "texture" that can mask animal signatures.
- The best detection conditions are **early morning (1–3 hours after sunrise)** when the canopy surface is still cool from the night and animals are warm relative to their environment.

The Optimal Survey Window

Time of day	Detection conditions	Reason
Pre-dawn (04:00–06:00)	Excellent	Maximum thermal contrast — animals are warmest relative to near-freezing ambient at altitude
Early morning (06:00–09:00)	Good	High contrast. Canopy still cool. Most diurnal species active and exposed.
Late morning (09:00–11:00)	Acceptable	Solar heating reduces contrast. Still detectable for medium-large endotherms.
Midday (11:00–14:00)	Poor	Canopy surface approaches animal body temperature. Detection probability severely reduced.
Afternoon (14:00–17:00)	Variable	Shaded areas cool; sun-facing canopy still warm. Depends heavily on cloud cover.
Post-sunset (18:00–20:00)	Good for nocturnal species	Rapid canopy cooling restores contrast. Best for nocturnal primates and reptiles.

Altitude and Sensor Footprint

Altitude AGL	Footprint (Mavic 3T)	Pixel size (GSD)	Detection range
30 m	~18 × 13 m	~1.6 cm	Small mammals, large reptiles detectable
50 m	~30 × 22 m	~2.7 cm	Medium mammals reliably detectable
80 m	~48 × 35 m	~4.3 cm	Large mammals reliable. Small mammals borderline.
120 m	~73 × 53 m	~6.5 cm	Only large mammals (Indri, fosa) reliable

GSD = Ground Sampling Distance. Footprint values are approximate for Mavic 3T FLIR Boson 640 sensor. Altitude choice is a tradeoff between area coverage rate and detection probability for target species.

Canopy Cover and Detection Probability

Canopy cover is the single strongest predictor of detection probability in forested thermal surveys. The relationship is roughly linear: each 10% increase in canopy cover reduces detection probability by approximately 15–25%, depending on canopy structure and species size.

THE CANOPY COVER CORRECTION — CRITICAL FOR ERIS SCORING

A detection rate of 5 animals per km² in open canopy (20% cover) is not comparable to a detection rate of 5 animals per km² in dense canopy (80% cover). The dense canopy site almost certainly has a higher true density — the lower detection rate reflects lower detectability, not lower abundance.

ERIS thermal habitat quality scoring accounts for this by converting raw detection rates into corrected density estimates using canopy cover as a detection probability covariate. The detection data and the canopy cover estimate must both be recorded for this correction to be applied.

Flight Planning for Data Quality

Flight parameters directly determine data quality. These decisions must be made before departure — not adjusted in the field without documentation.

Systematic vs. Targeted Survey Design

Design	Description	Best for	Limitation
Systematic transect grid	Parallel flight lines at fixed spacing covering the study area	Abundance estimation, detection probability modelling, ERIS habitat quality scoring	Time-intensive. Large areas require multiple flights.
Habitat-targeted transects	Flight lines placed over specific habitat types (stream edges, fruiting trees, ridge lines)	Species-specific surveys. Maximises detection rate for known habitat associations.	Not suitable for unbiased abundance estimation. Requires prior habitat knowledge.
Fixed-point hovering	Aircraft hovers at fixed points for 2–3 minutes watching for animal movement	Group size estimation at known aggregation sites (e.g., known lemur home ranges)	Very small area coverage. Not scalable for biodiversity surveys.

Recommended Flight Parameters — Mavic 3T (ERIS standard)

Mavic 3T — Standard ERIS Survey Protocol		Validated at Ihofa Forest, 2024–2025
Altitude AGL	60–80 m (70 m recommended for mixed forest). Reduces to 40 m over open canopy areas for small mammal detection.	
Flight speed	3–4 m/s. At 4 m/s with 1-second trigger: ~4m between frames. At 3 m/s: ~3m between frames. Slower = more consecutive frames per detection = easier merging.	
Image trigger	Time-triggered: 1 image per second. Distance-triggered at 3m intervals is also acceptable but requires flight speed control.	
Transect spacing	For ≥80% coverage: transect spacing = $0.8 \times$ sensor footprint width at survey altitude. At 70m altitude: footprint ~38m → spacing ~30m.	
Survey window	05:30–09:00 local time for diurnal species. 19:30–22:00 for nocturnal species. Document start and end time for every flight.	
Thermal palette	Ironbow for all detection review. Temperature range: manual setting 15°C–40°C for tropical morning surveys. Auto-range acceptable for habitat mapping only.	
Simultaneous capture	Enable RGB camera simultaneous capture where possible. RGB images provide canopy cover estimation and help resolve false positives.	
Pre-flight records	Record: ambient temperature (°C), wind speed estimate (m/s), cloud cover (%), battery charge (%), flight start time (UTC and local). These become effort covariables.	

AI Prompt Pack — 12 Prompts

Prompt 1 — Flight Metadata Extraction Plan

I have [number] thermal image files from a drone survey using [camera model]. Help me design a batch metadata extraction workflow. Camera model: [e.g., DJI Mavic 3T / DJI XT2 / FLIR Vue Pro] File format: [RJPEG / TIFF / JPG] Operating system: [Windows / Mac / Linux] Available tools: [exiftool / Python / R / none yet] Generate: 1. The exiftool command to extract GPS, timestamp, altitude, and camera settings from all files in a folder into a CSV 2. The Python code equivalent using the Pillow or piexif library 3. A list of which EXIF tags contain the GPS coordinates for this specific camera model 4. Any known issues with metadata extraction for this camera model (e.g., DJI RJPEG proprietary encoding) 5. How to verify the extracted GPS coordinates against the DJI flight log (the .DAT or .SRT file)

Prompt 2 — False Positive Classification Guidance

I am reviewing thermal drone imagery from a survey over [habitat type] in [region]. I have flagged [number] candidate detections and need to systematically classify them as genuine wildlife signatures or false positives. Study context: - Target taxa: [species or groups] - Survey time: [time of day and ambient temperature] - Altitude: [metres AGL] - Habitat: [forest type and canopy cover estimate] - Platform: [drone model] For each of the following candidate signatures I am uncertain about, provide: 1. The most likely explanation (genuine detection or specific false positive type) 2. The diagnostic features I should look for to confirm 3. The action to take: RETAIN as detection / REJECT as false positive / FLAG for second review Signatures to classify: [Describe each uncertain signature – shape, size, temperature relative to background, behaviour across frames, habitat context]

Prompt 3 — Consecutive-Frame Merge Script

Generate a Python script to identify consecutive-frame detection groups from my thermal review log. Input: A CSV file with columns: frame_filename, frame_number, detection_flagged (TRUE/FALSE), GPS_lat, GPS_lon Logic: - Group consecutive flagged frames where the gap between frame numbers is $\leq [3]$ frames - For each group: calculate the GPS centroid of all frames in the group - For each group: identify the frame closest to the group GPS centroid as the "representative frame" - Assign a unique detection_id to each group - Output: one row per detection group with: detection_id, start_frame, end_frame, n_frames, centroid_lat, centroid_lon, representative_frame, GPS_uncertainty_m GPS uncertainty estimation: for Mavic 3T at [altitude]m altitude with [n] frames per detection, estimate positional uncertainty as [formula based on altitude and GPS accuracy]. Output format: CSV with one row per merged detection.

Prompt 4 — Species Identification from Thermal Signature

I am trying to classify a thermal detection from a drone survey in [region] at [altitude]m altitude. Help me narrow down the species identification. Detection characteristics: - Estimated body length (from pixel count $\times$ GSD): [cm] - Estimated body width: [cm] - Shape: [elongated / compact / irregular] - Behaviour across frames: [stationary / slow movement / fast movement / climbing / in canopy / on ground] - Location in frame: [canopy / edge / ground / branch / tree trunk] - Time of survey: [time and ambient temperature] - Number of individuals in same frame: [number] Known species for this habitat: [List likely species with approximate body sizes] Classify the detection as: 1. Most likely species, with reasoning 2. Alternative species to consider 3. Recommended confidence level (HIGH / MEDIUM / LOW / UNIDENTIFIED) 4. What additional data (audio, RGB image, movement pattern) would upgrade the confidence level

Prompt 5 — YOLO Training Data Preparation

I want to train or fine-tune a YOLO wildlife detection model on my thermal drone imagery. Help me plan the training data preparation. My dataset: - Total thermal images: [number] - Validated detections (manual review): [number] - Target species: [list] - Image format: [RJPEG / TIFF / PNG] - Image dimensions: [pixels] - Platform used for annotation: [CVAT / LabelImg / Roboflow / none yet] Generate: 1. Minimum training dataset requirements for the target species and detection frequency 2. Recommended train/validation/test split for this dataset size 3. Annotation format required for YOLO (YOLOv8 / v9) – bounding box format and class labels 4. Data augmentation strategies specifically appropriate for thermal wildlife imagery 5. Transfer learning recommendation: which base model to fine-tune from (FLIR dataset / COCO / custom) 6. Evaluation metrics to use: mAP threshold, per-class recall, false positive rate target for wildlife detection

Prompt 6 — Detection Confidence Scoring Rules

Help me write a species classification confidence scoring guide for thermal drone detections in my study area. Study area: [region and habitat type] Target species list: [list species with approximate body mass and thermal characteristics] Survey platform: [drone model and altitude] Survey time window: [time and ambient temperature range] For each species on my target list, define: 1. The specific criteria that would justify HIGH confidence classification from thermal data alone 2. The criteria for MEDIUM confidence (consistent with species but not diagnostic) 3. What additional data (audio call, RGB image, behaviour) would upgrade from MEDIUM to HIGH 4. The minimum body temperature differential from background that enables detection at survey altitude 5. A simple decision tree: what to look for first, second, third when classifying a detection as this species Format: a reference card I can print and use during image review sessions.

Prompt 7 — Habitat Quality Thermal Map Interpretation

I want to extract habitat quality metrics from the thermal imagery of my survey area (not just detections – the thermal profile of the habitat itself). Survey context: - Study area size: [hectares] - Survey time: [time and season] - Platform: [drone model and altitude] - Target habitat metrics: [canopy cover / thermal heterogeneity / heat island detection / water feature mapping / etc.] For the thermal mosaic of my survey area, help me: 1. Define canopy cover estimation from thermal imagery (proportion of frame below a threshold temperature) 2. Define a thermal heterogeneity index (variance in pixel temperature across the mosaic) 3. Identify water features (streams, wetlands) from thermal signatures 4. Identify edge habitats where thermal contrast is highest (and detection probability is highest) 5. Generate a scoring rubric for "thermal habitat quality" that could contribute to an ERIS biodiversity score Provide Python code using OpenCV or rasterio to calculate each metric from a thermal GeoTIFF.

Prompt 8 — ERIS Detection Record Schema Converter

I have a thermal wildlife detection dataset from my drone survey. Convert it to the ERIS detection record schema. My current dataset columns: [List your current column names and types] ERIS target schema (from the OPLURIX Biodiversity Data Harmonizer): - record_id, survey_method (THERMAL_DRONE), detection_date, detection_time, site_code, station_id, latitude, longitude, species_code, species_latin, count, detection_type (COUNT), confidence, thermal_signature_id, source_frames, flight_id, altitude_m, ambient_temp_c, canopy_cover_pct, gps_accuracy_m, sensitivity_flag Generate: 1. A column mapping table: my column → ERIS column 2. Any ERIS columns that are missing from my dataset and how to derive or estimate them 3. Python code to perform the conversion 4. Validation checks to run on the converted dataset before ERIS submission

Prompt 9 — Inter-Observer Calibration Protocol

I need to calibrate two or more reviewers who will be reviewing thermal imagery independently. Design an inter-observer calibration protocol. Context: - Number of reviewers: [number] - Reviewer experience: [expert / intermediate / novice] - Target species: [list] - Estimated total images to review: [number] - Time available for calibration: [hours] Generate: 1. A calibration image set size recommendation (number of images for training review) 2. A standardised review protocol that all reviewers will follow 3. A method for calculating inter-observer agreement (Cohen's kappa or simpler percentage agreement) 4. A threshold for acceptable agreement before full dataset review begins 5. A procedure for resolving disagreements on disputed detections 6. A form for tracking reviewer performance across the review session

Prompt 10 — EXIF Batch Extraction Script

Generate a Python script to batch-extract GPS and flight metadata from DJI thermal image files. Input: A folder of DJI [Mavic 3T / XT2 / H30T] thermal image files in [RJPEG / TIFF / JPG] format. Extract for each file: - Filename - GPS latitude (decimal degrees, WGS84) - GPS longitude (decimal degrees, WGS84) - GPS altitude (metres above sea level, from EXIF) - Relative altitude AGL (from DJI EXIF if available) - Image timestamp (UTC) - Camera model - Any DJI-specific flight telemetry fields available in the EXIF Output: A CSV file with one row per image. Requirements: - Use exiftool via subprocess (most reliable for DJI RJPEG files) OR use the exifread Python library - Handle missing EXIF fields gracefully (output NA rather than crashing) - Flag any images where GPS coordinates are (0,0) or clearly invalid - Include a summary: total images processed, images with valid GPS, images with missing GPS

Prompt 11 — Thermal Mosaicking and Georeferencing Guide

I need to create a georeferenced thermal mosaic from my drone survey images for habitat quality mapping. Survey details: - Platform: [drone model] - Total images: [number] - Overlap: [forward overlap %] × [side overlap %] - GPS accuracy: [estimate in metres] - GCP (Ground Control Points) available: [yes/no] - Software available: [Agisoft Metashape / WebODM / Pix4D / none] Generate: 1. Recommended software and settings for processing thermal drone imagery into an orthomosaic 2. Key differences between processing thermal and RGB imagery (important – settings differ significantly) 3. Whether GCPs are necessary at this GPS accuracy level for the intended analysis 4. Output format recommendations for raster analysis in QGIS or Python (GeoTIFF with temperature values preserved) 5. Validation checks to verify the mosaic quality before using it for analysis

Prompt 12 — Detection Report Generator

Generate a flight-level detection summary report for my thermal survey. Flight details: - Flight ID: [ID] - Date and time: [datetime] - Site: [site name] - Area covered: [hectares] - Survey duration: [minutes] - Altitude: [metres] - Ambient temperature: [°C] - Canopy cover estimate: [%] Detection results: - Total images reviewed: [number] - Images with candidate detections: [number] - False positives removed: [number] (breakdown by type if available) - Validated detections: [number] - Unique individuals (after consecutive-frame merging): [number] - By species/confidence level: [list] Generate: 1. A one-page flight report suitable for the mission debrief 2. A raw detection rate and effort-corrected detection rate (detections per km² surveyed) 3. Notes on detection conditions (good / fair / poor) based on time and temperature 4. Flags for any methodological issues encountered (equipment problems, flight deviations, weather) 5. Recommended follow-up surveys if detection rate was below expected

Species Detection Guide — Madagascar Thermal Surveys

Thermal signature characteristics and detection criteria for the most commonly encountered species in Madagascar forest thermal drone surveys. Based on fieldwork at Ihofa, Kirindy, and Beza Mahafaly. All size estimates are at 70m altitude using Mavic 3T FLIR Boson sensor (GSD ~4 cm/pixel).

Primates

Species	Body mass	Pixel size at 70m	Thermal signature	Detection criteria	Confidence achievable
<i>Indri indri</i>	6–9 kg	~15×10 px	Bright, compact, bilobed (head + body). One of the most detectable species. Often calling — audio confirmation easy.	Size + shape + habitat (upper canopy). Audio confirmation raises to HIGH.	HIGH with audio; MEDIUM thermal only
<i>Propithecus</i> spp.	3–6 kg	~12×8 px	Similar to Indri but smaller and more elongated during clinging posture. Vertical tree trunks.	Size + vertical position. Species-level ID requires size comparison against Indri if both present.	MEDIUM (genus level often more honest)
<i>Hapalemur</i> spp.	0.7–2.5 kg	~6×4 px	Small, compact, near bamboo or grass edge. Easily confused with birds at 70m altitude.	Detection difficult at 70m. Reduce to 40m for reliable detection. Bamboo habitat context helps.	LOW at 70m; MEDIUM at 40m
<i>Lemur catta</i>	2–3 kg	~8×6 px	Compact, often in groups. Ground-foraging detectable in open areas. Long tail sometimes visible as trailing warm streak.	Group behaviour + semi-open habitat + correct size range.	MEDIUM–HIGH in open habitat
<i>Eulemur</i> spp.	2–3.5 kg	~9×6 px	Medium, often in pairs or small groups. Mid-canopy to upper canopy. Active at dawn.	Size + group size + time of detection. Species-level requires additional evidence.	LOW–MEDIUM (genus level)
<i>Microcebus</i> spp.	40–80 g	~2×2 px	Near limit of detection at 70m. 1–2 pixel warm dots. Not reliably detectable above 40m altitude.	Only detect below 40m AGL. Cannot distinguish from large insects at altitude.	UNIDENTIFIED above 40m

Carnivores

Species	Body mass	Thermal signature	Detection notes
<i>Cryptoprocta ferox</i> (fosa)	5–10 kg	Large, elongated, single bright signature. Most often ground-level or low branches at dawn/dusk. Moves steadily along trails.	Most detectable carnivore in thermal surveys. Track or road following behaviour makes detection easier. HIGH confidence achievable if size + habitat + behaviour consistent.
<i>Fossa fossana</i>	1.5–2 kg	Small elongated signature. Ground level. Nocturnal.	Detectable at 40–50m. Easily confused with small lemurs at 70m. LOW confidence at high altitude.

Common False Positives at This Study Site

Source	Thermal appearance	Distinguishing feature
Termite mounds (Macrotermes)	Very bright, circular, much hotter than background	Temperature clipping in Ironbow palette (solid white centre). Stationary across days. Common in dry forest.
Exposed rock outcrops	Warm, irregular shape, large area	Matches topography visible in RGB. No movement. Consistent across reflights.
Large birds (raptors, ground birds)	Small compact signature with rapid movement	Movement speed too fast for a mammal. Often airborne — appears in one frame only. No consistent flight path correlation with transect direction.
Chameleons (on exposed branch)	Very small, elongated, on branch surface	Extremely slow movement or stationary. Ectotherm — temperature tracks ambient closely, reducing contrast by midday. Good contrast at dawn if recently emerged from cool roost.

Code Templates

Python — EXIF Batch Extraction from DJI Thermal Files

PYTHON · EXIFTOOL SUBPROCESS · DJI THERMAL EXIF EXTRACTION

```
import subprocess, csv, os, json
def extract_thermal_exif(folder_path: str, output_csv: str) -> None:
    """Extract GPS and flight metadata from all thermal files in folder."""
    cmd = [ 'exiftool', '-r', '-json', '-GPSLatitude#', '-GPSLongitude#', '-GPSAltitude#', '-DateTimeOriginal', '-RelativeAltitude', '-Model', '-FileName', folder_path ]
    result = subprocess.run(cmd, capture_output=True, text=True)
    records = json.loads(result.stdout)
    fields = ['FileName', 'DateTimeOriginal', 'GPSLatitude', 'GPSLongitude', 'GPSAltitude', 'RelativeAltitude', 'Model']
    with open(output_csv, 'w', newline='') as f:
        writer = csv.DictWriter(f, fieldnames=fields + ['gps_flag'])
        writer.writeheader()
        for r in records:
            lat = r.get('GPSLatitude')
            lon = r.get('GPSLongitude')
            flag = 'VALID'
            if lat is None or lon is None:
                flag = 'MISSING_GPS'
            elif abs(lat) < 0.001 and abs(lon) < 0.001:
                flag = 'NULL_ISLAND'
            row = {k: r.get(k, 'NA') for k in fields}
            row['gps_flag'] = flag
            writer.writerow(row)
            print(f"Extracted {len(records)} records -> {output_csv}")
```

Python — Consecutive-Frame Detection Merger

PYTHON · PANDAS · MERGE CONSECUTIVE FLAGGED FRAMES INTO SINGLE DETECTIONS

```
import pandas as pd
import numpy as np
def merge_detections(df: pd.DataFrame, max_gap: int = 3) -> pd.DataFrame:
    """Group consecutive flagged frames into single detection records."""
    flagged = df[df['detection_flagged']].sort_values('frame_number').copy()
    flagged['group'] = (flagged['frame_number'].diff() > max_gap).cumsum()
    groups = []
    for gid, grp in flagged.groupby('group'):
        best_idx = ((grp['GPS_lat'] - grp['GPS_lat'].mean())**2 + (grp['GPS_lon'] - grp['GPS_lon'].mean())**2).idxmin()
        groups.append({
            'detection_id': f"DET_{gid:04d}",
            'start_frame': grp['frame_number'].min(),
            'end_frame': grp['frame_number'].max(),
            'n_frames': len(grp),
            'latitude': round(grp.loc[best_idx, 'GPS_lat'], 5),
            'longitude': round(grp.loc[best_idx, 'GPS_lon'], 5),
            'source_frames': ';'.join(grp['frame_filename'].tolist())
        })
    return pd.DataFrame(groups)
```

R — Canopy Cover Estimation from Thermal Orthomosaic

```
R · TERRA · CANOPY COVER FROM THERMAL TEMPERATURE RASTER
```

```
library(terra) # Load thermal orthomosaic (temperature values in Kelvin or Celsius) thermal
<- rast("IH0FA_20260315_thermal_ortho.tif") # Canopy surface temperature threshold: # Below
this = canopy (shaded/cool leaves) # Above this = gap / ground / animal # Calibrate from
your specific survey conditions ambient_temp_c <- 22 # measured before flight
canopy_threshold <- ambient_temp_c + 2 # 2°C above ambient = canopy surface # Binary canopy
mask: 1 = canopy, 0 = gap or warm surface canopy_mask <- ifel(thermal <= canopy_threshold,
1, 0) # Canopy cover per 100m grid cell grid_100m <- aggregate(canopy_mask, fact = 100, fun
= "mean") names(grid_100m) <- "canopy_cover_pct" grid_100m["canopy_cover_pct"] <-
grid_100m["canopy_cover_pct"] * 100 writeRaster(grid_100m, "canopy_cover_100m.tif",
overwrite = TRUE)
```

Thermal Detection Record Schema

The complete output schema for a thermal wildlife detection dataset. This schema is compatible with the OPLURIX Biodiversity Data Harmonizer (Product 7) and the ERIS biodiversity scoring input format.

```

THERMAL DETECTION RECORD – MASTER SCHEMA v1.0 Compatible with: OPLURIX Data Harmonizer ·
ERIS Scoring Input · Darwin Core PROVENANCE COLUMNS (required in all records) record_id
TEXT Unique. Format: [SITE]_[DATE]_[FLIGHT]_[DET_NUM] survey_method CODE THERMAL_DRONE
(always for this schema) source_file_id TEXT Flight folder ID from dataset inventory
flight_id TEXT Unique flight identifier merge_date DATE Date this record entered the
dataset DETECTION COLUMNS detection_date DATE YYYY-MM-DD detection_time TIME HH:MM (local
time) site_code CODE From Site Registry station_id TEXT Transect pass ID (e.g., T01_PASS2)
latitude LAT Decimal degrees WGS84, 5 d.p. longitude LON Decimal degrees WGS84, 5 d.p.
gps_accuracy_m FLOAT Estimated GPS accuracy in metres coord_original TEXT Raw GPS from EXIF
(preserve original) spatial_flag FLAG VALID / SPATIAL_OUTLIER / LOW_PRECISION / MISSING
SPECIES COLUMNS species_code CODE From Species Registry. WARM_SIG_MAMMAL if unidentified.
species_latin TEXT Full binomial or "cf." or family species_raw TEXT Reviewer's original
annotation confidence FLAG HIGH / MEDIUM / LOW / UNIDENTIFIED classification_basis TEXT
THERMAL_SIZE / THERMAL_BEHAVIOUR / AUDIO / RGB / YOLO_MODEL DETECTION METADATA count INT
Number of individuals (after consecutive-frame merging) source_frames TEXT Semicolon-
separated list of source image filenames n_source_frames INT Number of frames this
detection spans detection_type CODE COUNT (always for thermal detections)
false_positive_check FLAG VERIFIED / FP_REMOVED (for records removed from main dataset)
EFFORT COVARIABLES (required for detection probability modelling) altitude_m FLOAT Flight
altitude above ground level in metres ambient_temp_c FLOAT Ambient temperature at survey
time in °C canopy_cover_pct FLOAT Estimated canopy cover at detection location (0-100)
wind_speed_ms FLOAT Wind speed in m/s (estimate acceptable) survey_start_time TIME HH:MM
UTC thermal_palette CODE IRONBOW / RAINBOW / GRAYSCALE / AUTO SENSITIVITY sensitivity_flag
FLAG PUBLIC / RESTRICTED / OFFSET_APPLIED QUALITY reviewer_code CODE Observer code of image
reviewer review_date DATE Date of image review qc_pass BOOL TRUE if passed three-check QC
framework notes TEXT Free text for any record-level issues

```

ERIS Integration — Thermal Habitat Quality Scoring

How thermal drone data contributes to the ERIS ecological integrity score. The thermal survey produces two ERIS indicator inputs: detection-based species richness data and thermal habitat quality metrics.

ERIS Indicator 3 — Thermal Habitat Quality

This indicator characterises the thermal structure of the survey area — the distribution of microhabitat temperatures, canopy cover, and thermal heterogeneity that determine habitat suitability for target species.

ERIS THERMAL HABITAT QUALITY — FOUR SUB-METRICS

Mean canopy cover across the survey area from the thermal orthomosaic (%). Higher closed-canopy cover = higher forest integrity score for forest-dependent species.

Variance in canopy surface temperature across the survey area. High heterogeneity = greater microhabitat diversity = higher habitat quality for thermally sensitive species (amphibians, reptiles).

Linear metres of thermal contrast edge (forest-to-gap boundaries) per hectare. High edge density can indicate fragmentation or degradation.

Detections per km² of thermal survey area, corrected for canopy cover using the detection probability model. This connects the thermal detection data to the ERIS species richness indicator.

From Thermal Data to ERIS Score — the Calculation Path

1. **Complete the 7-step processing pipeline** → validated detection dataset + flight effort table
2. **Generate the thermal orthomosaic** (if multiple overlapping passes) or use individual images for single-pass surveys → thermal habitat raster
3. **Calculate canopy cover, thermal heterogeneity, and edge density** from the raster using the R code template (Section 9)
4. **Calculate detection-corrected wildlife density** using canopy cover as a detection probability covariate (distance sampling or N-mixture model)
5. **Input all four sub-metrics into the ERIS scoring framework** using the Data Harmonizer output schema

SECTION 12

Three-Check Quality Framework

Check 1 — False Positive Rate Audit

WHAT IT TESTS

What percentage of initial detections were removed as false positives? A rate above 40% indicates systematic problems with survey conditions or review protocol.

Pass criteria: False positive rate documented and below 40%. Each false positive category is logged.

Check 2 — Inter-Observer Agreement

WHAT IT TESTS

If two reviewers independently reviewed the same image subset, how closely do their detections agree? Below 70% agreement indicates that the review protocol needs clarification before full dataset review.

Pass criteria: $\geq 70\%$ agreement on detection presence/absence. $\geq 60\%$ agreement on species confidence classification.

Check 3 — GPS Validity and Coverage

WHAT IT TESTS

Are all detection GPS coordinates within the study area? Does the spatial distribution of detections match the flight path? (Detections that cluster far from the flight path indicate GPS error.)

Pass criteria: $\geq 95\%$ of detections within study area bounding box. Detection spatial distribution consistent with flight path within GPS accuracy estimate.

Quick Start — Process Your First Flight

Before You Begin

- ☐ Confirm all thermal image files are copied from the memory card to working storage. Do not process from the card.
- ☐ Record the flight metadata: site, date, time, altitude, ambient temperature, canopy cover estimate, pilot.
- ☐ Install `exiftool` (command-line) and Python with `pandas` , `pillow` .

Hours 1–2 — File Organisation and Extraction

- ☐ Create a flight folder with the correct naming convention.
- ☐ Run the EXIF extraction script (Section 9, Prompt 10). Verify GPS coordinates for a sample of images on a map.
- ☐ Set all images to Ironbow palette with consistent temperature range. Note the range used.

Hours 3–4 — Detection Review and False Positive Removal

- ☐ Fast-scroll through all images at 4–6 fps. Flag images with candidate detections.
- ☐ For each flagged image: apply the false positive decision framework (Section 5). Log every removal.
- ☐ Record: total images, images flagged, false positives removed, false positive rate.

Hour 5 — Consecutive-Frame Merging and Classification

- ☐ Run the consecutive-frame merge script (Section 9) on the validated flagged list.
- ☐ For each merged detection: classify species and assign confidence level using Species Detection Guide (Section 8).
- ☐ Populate all required columns of the detection record schema (Section 10).

Hour 6 — Validation and Documentation

- ☐ Run the three-check QC framework (Section 12). Document results.
- ☐ Use Prompt 12 to generate the flight detection report.
- ☐ Run the ERIS schema converter (Prompt 8) if submitting to ERIS.
- ☐ Back up the processed dataset to two locations.

Minimum Viable vs. Premium Version

Feature	This Version (MVP)	Premium (Future)
7-step processing pipeline	✓	✓
Thermal image interpretation guide	✓	✓ + visual examples
Flight planning for data quality	✓	✓
12 AI prompts	✓	✓ 20 prompts
Species detection guide (Madagascar)	✓	✓ + East/West Africa
Python/R code templates (5)	✓	✓ 12 templates
Detection record schema	✓	✓ ERIS v2 compatible
ERIS thermal habitat scoring guide	✓	✓ + scoring calculator
Three-check QC framework	✓	✓
Pre-trained YOLO model (Madagascar lemurs)	—	Planned (requires dataset)
Automated false positive classifier	—	Planned
DJI Matrice 350 RTK + H30T specific guide	—	Planned (grant-dependent)
French version	—	Planned

A thermal image is not a detection. A processed, validated, documented *thermal detection record is.*

The gap between raw thermal footage and publishable biodiversity data is a workflow problem, not a technology problem. This guide closes that gap — and in doing so, connects every thermal survey you run to the ERIS ecological integrity infrastructure that will make conservation finance accessible to the forests you work in.

Thermal Data Translator v1.0

An OPLURIX Product · 2026

Proof of concept: IPS Congress 2025 — Thermal Drone Lemur Surveys

[marvelous-Valkyrie-f94b2f.netlify.app](https://marvelous- Valkyrie-f94b2f.netlify.app)

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